```
import pandas as pd

clients = pd.read_csv('clients.csv')
properties = pd.read_csv('properties.csv')

clients.head()
```

	client_id	client_type	first_name	last_name	date_of_birth	gender	country	region	ac
0	C0001	Individual	Kareem	Liu	05-11-1968	F	USA	California	
1	C0002	Individual	Trystan	Oconnor	11/26/1962	M	USA	California	
2	C0003	Individual	Kale	Gay	04-07-1959	M	USA	California	
3	C0004	Individual	Russell	Gross	11/25/1959	M	USA	California	
4	C0005	Company	Marleez	Co	2/28/1976	M	USA	California	

Next steps: Generate code with `clients` New interactive sheet

```
clients.columns
```

```
Index(['client_id', 'client_type', 'first_name', 'last_name', 'date_of_birth',
       'gender', 'country', 'region', 'acquisition_purpose',
       'satisfaction_score', 'loan_applied', 'referral_channel'],
      dtype='object')
```

```
clients.isnull().sum()
```

	0
client_id	0
client_type	0
first_name	0
last_name	0
date_of_birth	0
gender	0
country	0
region	0
acquisition_purpose	0
satisfaction_score	0
loan_applied	0
referral_channel	0

dtype: int64

```
clients = clients.drop_duplicates()
```

```
clients = clients.fillna(method='ffill')
```

```
/tmp/ipykernel_3152/832310432.py:1: FutureWarning: DataFrame.fillna with 'method' is deprecat
  clients = clients.fillna(method='ffill')
```

```
clients.isnull().sum()
```

	0
client_id	0
client_type	0
first_name	0
last_name	0
date_of_birth	0
gender	0
country	0
region	0
acquisition_purpose	0
satisfaction_score	0
loan_applied	0
referral_channel	0

dtype: int64

```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()

clients['client_type'] = le.fit_transform(clients['client_type'])
clients['acquisition_purpose'] = le.fit_transform(clients['acquisition_purpose'])
clients['region'] = le.fit_transform(clients['region'])
clients['country'] = le.fit_transform(clients['country'])
clients['referral_channel'] = le.fit_transform(clients['referral_channel'])

clients.head()
```

	client_id	client_type	first_name	last_name	date_of_birth	gender	country	region	acqu
0	C0001	1	Kareem	Liu	05-11-1968	F	9	8	
1	C0002	1	Trystan	Oconnor	11/26/1962	M	9	8	
2	C0003	1	Kale	Gay	04-07-1959	M	9	8	
3	C0004	1	Russell	Gross	11/25/1959	M	9	8	
4	C0005	0	Marleez	Co	2/28/1976	M	9	8	

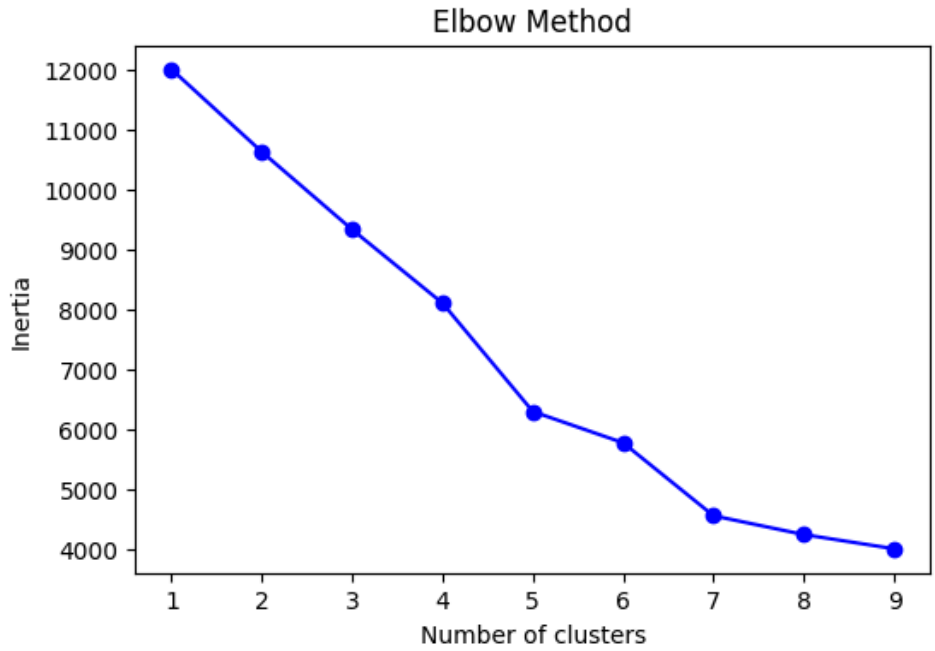
Next steps: Generate code with clients New interactive sheet

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

scaled_data = scaler.fit_transform(clients.select_dtypes(include=['int64','float64']))
```

```
from sklearn.cluster import KMeans
```

```
import matplotlib.pyplot as plt
import numpy as np
```

```
# Elbow method
inertia = []
K = range(1,10)
for k in K:
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(scaled_data)
    inertia.append(kmeans.inertia_)

plt.figure(figsize=(6,4))
plt.plot(K, inertia, 'bo-')
plt.xlabel('Number of clusters')
plt.ylabel('Inertia')
plt.title('Elbow Method')
plt.show()
```




```
kmeans = KMeans(n_clusters=4, random_state=42)
clients['cluster'] = kmeans.fit_predict(scaled_data)

clients['cluster'].value_counts()
```

cluster	count
0	606
3	589
1	419
2	386

dtype: int64

```
numeric_cols = clients.select_dtypes(include=['int64','float64']).columns
cluster_summary = clients.groupby('cluster')[numeric_cols].mean()
cluster_summary
```

cluster	client_type	country	region	acquisition_purpose	satisfaction_score	referral_c
0	0.942244	7.985149	20.391089	1.00000	3.036304	1.
1	0.959427	6.935561	43.990453	0.02148	3.124105	1.
2	0.958549	8.375648	11.466321	0.00000	4.531088	1.
3	0.940577	8.261460	12.315789	0.00000	1.969440	1.

Next steps: Generate code with `cluster_summary` New interactive sheet

```
!pip install streamlit --quiet

import streamlit as st
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

st.title("Buyer Segmentation Dashboard - Parcl Real Estate")

st.write("This dashboard shows buyer clusters and their investment patterns.")

st.subheader("Cluster Distribution")

cluster_counts = clients['cluster'].value_counts()
st.bar_chart(cluster_counts)

st.subheader("Investment Patterns by Cluster")

fig, ax = plt.subplots(figsize=(8,4))
sns.countplot(x='cluster', hue='acquisition_purpose', data=clients, ax=ax)
st.pyplot(fig)

st.subheader("Geographic Buyer Analysis")

fig, ax = plt.subplots(figsize=(8,4))
sns.countplot(x='region', hue='cluster', data=clients, ax=ax)
st.pyplot(fig)

st.subheader("Cluster Insights (Average Numeric Features)")

numeric_cols = clients.select_dtypes(include=['int64','float64']).columns
cluster_summary = clients.groupby('cluster')[numeric_cols].mean()
st.dataframe(cluster_summary)

!streamlit run your_notebook_name.py
```

```
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2026-04-02 07:59:48.078
  Warning: to view this Streamlit app on a browser, run it with the following
  command:

    streamlit run /usr/local/lib/python3.12/dist-packages/colab_kernel_launcher.py [ARGUMEN
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2026-04-02 07:59:50.476 Thread 'MainThread': missing ScriptRunContext! This warning can be
2026-04-02 07:59:50.477 Thread 'MainThread': missing ScriptRunContext! This warning can be
2026-04-02 07:59:50.478 Thread 'MainThread': missing ScriptRunContext! This warning can be
```